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Clocks, Renaissance

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Abstract

The reintroduction of time-keeping technology to the Latin West from the thirteenth century and its subsequent submission to new applications produced the appearance of both large and small-scale clocks in the Renaissance. The great bulk of these works often replicated principles preserved from Classical antiquity and Medieval Arabian sources, but major technological advances appeared intermittently in the Renaissance which propelled clock-making technology forward. Just as we would no sooner exchange the precision of an atomic clock today for a sand-glass, so too in the Renaissance did old ways of clock-making rapidly become obsolete with the invention of new systems: weights, spring-drivers, wheels, gears, and organ-barrels. A similar phenomenon comparable to modern trends also has been observed with the Renaissance explosion of technology; as it became more advanced, it became smaller and more compact. What had been housed in the monumental proportions of Medieval Cathedral clock-towers had by the late Renaissance been reduced at its most

extreme to the size of a ring to be worn on an emperor's finger. The Renaissance aim of mastering the natural world around, space as well as time, can be inferred in the increasingly masterful method devised to standardize and measure the passage of not only the sun but the moon, the planets, the zodiac, and other visible astronomical phenomena as well.

Synonyms

Automation; Clockwork mechanics; Time-keeping technology

Innovative and Original Aspects

The time-keeping device, whatever its operating principle – sun, water, weights, springs, or gears – had stood in the culture of Classical antiquity inherited by the Renaissance humanists as the quintessential symbol of the ordered cosmos. In scholarship of the early-modern period, the clock became a seminal image for how many Enlightenment-era philosophers began to reimagine the cosmos and man's place within. In the period directly preceding this flourishing of thought, the thirteenth through the mid-seventeenth centuries, the development of time-keeping technology mirrored what was on a macroscale humanity's ever-increasing command and quantification of the surrounding natural world.

Technologically and artistically, the Renaissance continued to develop and refine a phenomenon most properly traced to the Medieval Period, as returning Crusaders and invigorated trade routes are credited with the reintroduction of time-keeping technology to Western Europe. The line of direct descent from Classical (predominantly derived from the work of Ctesibius and Hero of Alexandria) sources of clocks in Western Europe is believed to have been severed sometime after the documentation of a sixth-century clepsydra by Cassiodorus. That technology would not reappear until it had been filtered through the cultural influence of the Medieval Muslim East, through the preservation of Classical technology and subsequent original elaborations and novel applications, such as the dimension of human comfort, previously absent from known Greco-Roman mechanical devices, but prominent in the ideation of the robotic serving-girl in the *Book of Knowledge of Ingenious Devices* by Al-Jazari. Many mechanical arrangements observable in Arabic-engineering works have had to varying extents their corollaries located in existing medieval mechanical treatises. The water-clock, with its origins developed into the “turret clocks,” which substituted solid weights in lieu of water-weight. The most monumental assimilation of this reinvigorated mechanical and technological knowledge was put on display in the clock-towers of the great medieval Cathedrals (Strasbourg and Venice are early, large-scale examples of astrological clocks) built from the thirteenth century to the seventeenth, and in some cases beyond. As clocks became household objects and correspondingly diminished in size to a comfortable portability, they increased proportionally in prestige and esteem as objects of luxury and social status in Renaissance courtly and educated circles. The development of spring-technology in the 1400s facilitated the ever-increasing intricacy of minute clockwork, primarily in the German craft centers of Augsburg, Nuremberg, Dresden, and Ulm. Works created by the Habrecht brothers in the 1550s have been recognized as the earliest documented departures from strictly Heronic components: wheel-work replaced levers, gears replaced strings, and

organ-barrel programming replaced hydraulically delayed sequences. Clockwork technology began to infringe upon the monopoly of previously hydraulic-powered organs in princely gardens and some churches of the same time period. In the majority of cases, time-keeping technology was further ornamented with and embellished by mechanically driven moving automata, from the large cathedral-jacquemarts (bell-ringers) and assortment of angels, shepherds, and other major characters from predominantly (but not always) Biblical sources in religious and secular theatrical tableaux to the smaller automated minutiae of the opulently crafted mechanical *nefs* which propelled themselves around the tables of Rudolph II and other European rulers in the sixteenth and seventeenth centuries. In the case of the former, the prevailing inclination to veil the mechanical operations of mechanical devils, angels, animals, and humans in church dramas and elsewhere was intermittently lifted by engineers such as Giovanni Fontana, who wrote openly of the purely mechanical processes causing statues to spit fire or move their eyes and wings. A fifteenth/sixteenth-century automaton of a chained devil in the Collezione Permanente of MUDEC Milano provides perhaps the only surviving example in size, function, and style of the automata whose workings Fontana’s manuscript drawings reveal. The sixteenth-century automata of Juanelo Turriano, a lute-player in the Kunsthistorisches of Vienna and an automated monk currently in the Smithsonian’s collection, are miniaturized iterations of the larger theatre-automata powered by spring-driven clockwork and belong with the *nefs* in the latter class of objects created for the personal delight of a single patron (in the Turriano automata’s case Emperor Charles V) rather than an audience to a spectacle with religious or entertainment aims. Fantastic tales of mechanical inventions which made use of the rapidly developing engineering facilities of the Renaissance range from those generally acknowledged to be apocryphal (the android of Albert the Great or the iron fly of Regiomontanus and Bishop Virgilius of Naples) do not necessarily fall too far from descriptions of documented

automata created by Leonardo da Vinci and Hans Bullman.

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